

GCRew_Flume_TidalSurvey_DOC_QAQC_202308

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##Setup - Change things here & write any notes

```
#identify section  
cat("Setup Information")
```

Setup Information

```
##### Run information - PLEASE CHANGE  
Date_Run = "08/25/2023" #Date that instrument was run  
Run_by = "Stephanie J. Wilson" #Instrument user  
Script_run_by = "Stephanie J. Wilson" #Code user  
run_notes = " The standard curve for this run was checked manually on the instrument. " #any notes for this run  
samples_pattern <- "Flume"  
  #samples_pattern <- "GCW" #use this instead of the line above if you have only one site code  
chks_name = "Chk_Std_" #what did you name your check standards?  
crm_name = "CRM|crm" #what did you name your CRMS?  
  
library(here)
```

Warning: package 'here' was built under R version 4.3.3

here() starts at S:/Biogeochemistry/People/Steph Wilson/Github/serc-lateral-flux

```
##### File Names - PLEASE CHANGE  
#file path and name for raw summary data file  
  #raw_file_name = "Raw Data/TOCTN_COMPASS_Synoptic_DIC_202305.txt"  
raw_file_name = c(here("Tidal_Surveys/Raw_Data/202308", "TOCT_Flume_TidalSurvey_Aug2023_Spring_DIC.txt"),  
  here("Tidal_Surveys/Raw_Data/202308", "TOCT_Flume_TidalSurvey_Aug2023_Neap_DIC.txt"))  
  
#file path and name for raw all peaks file  
  #raw_allpeaks_name = "Raw Data/TOCTN_COMPASS_Synoptic_DIC_202305_allpeaks.txt"  
  
#file path and name of processed data file  
  #processed_file_name = "Processed Data/COMPASS_SynopticCB_PW_Processed_DIC_202308.csv"  
processed_file_name = here("Tidal_Surveys/Processed_Data/202308", "Flume_TidalSurvey_DIC_Processed.csv")  
  
##### Log Files - PLEASE CHECK  
#downloaded metadata csv - downloaded from Google drive as csv for this year  
Raw_Metadata = here("Tidal_Surveys/Raw_Data/202308", "Tidal_Survey_202308_Metadata.csv")
```

```
#qaqc log file path for this year
# Log_path = "Raw Data/COMPASS_Synoptic_DIC_QAQClog_2023.csv"
```

```
##Set Up Code
```

Import Data Functions

Import Sample Data

```
## Import Sample Data
```

```
## New names:
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## # A tibble: 6 x 3
```

```
##   sample_name      ic_raw run_datetime
##   <chr>           <dbl> <chr>
## 1 Flume_20230815_GW-1  93.9 8/18/2023 5:35:34 PM
## 2 Flume_20230815_GW-2  71.1 8/18/2023 5:55:35 PM
## 3 Flume_20230815_GW-3  30.2 8/18/2023 6:09:56 PM
## 4 Flume_20230815_GW-4  66.8 8/18/2023 6:30:15 PM
## 5 Flume_20230815_GW-5  61.9 8/18/2023 6:45:09 PM
## 6 Flume_20230815_GW-6  61.1 8/18/2023 6:59:55 PM
```

Assessing Standard Curves - this was done manually on the TOC, not included here

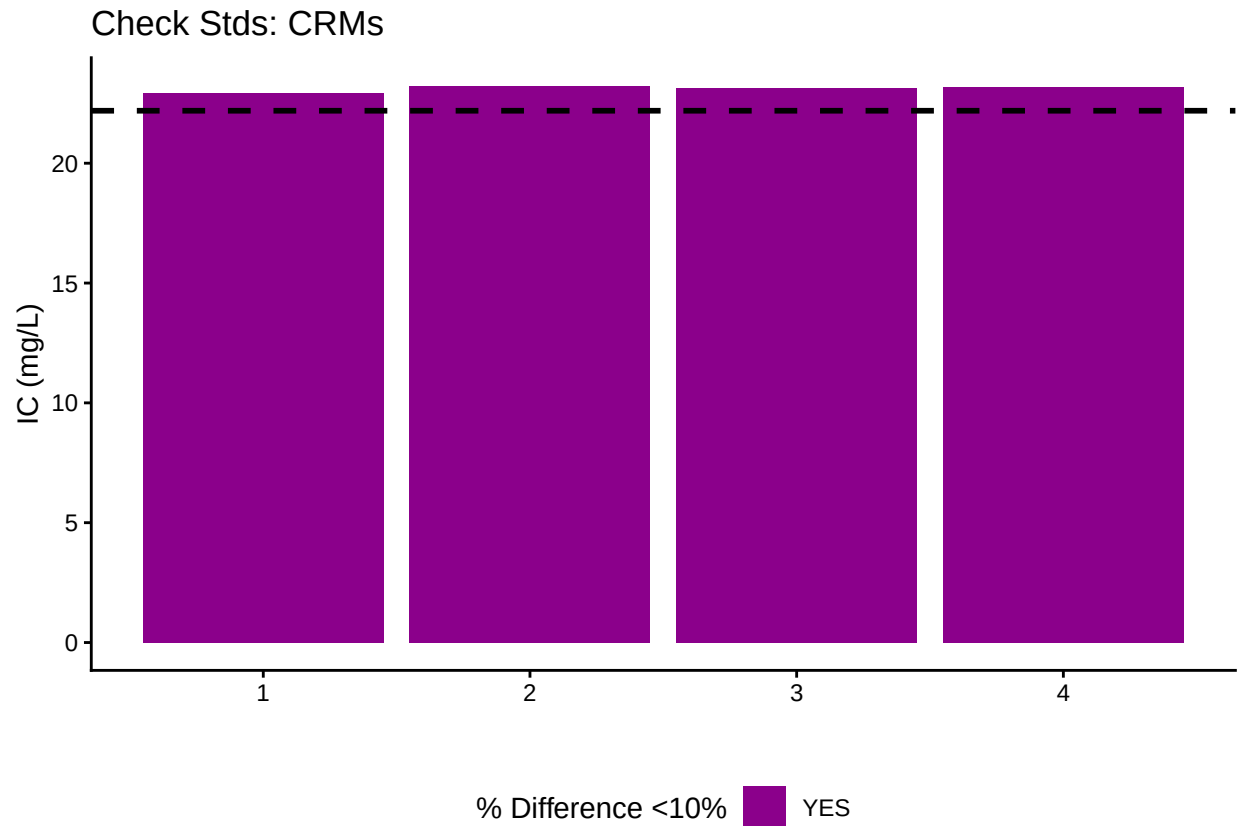
CRMs Check: CRM Batch 221-0866

```
## Assess the CRMs
```

```
## New names:
```

```
## New names:
```

```
## * ' ' -> '...14'
```



```
## [1] "IC crm has a % Difference <25% of expected - PROCEED"
```

```
## Run mean = 23.1175
```

```
## Expected = 22.19
```

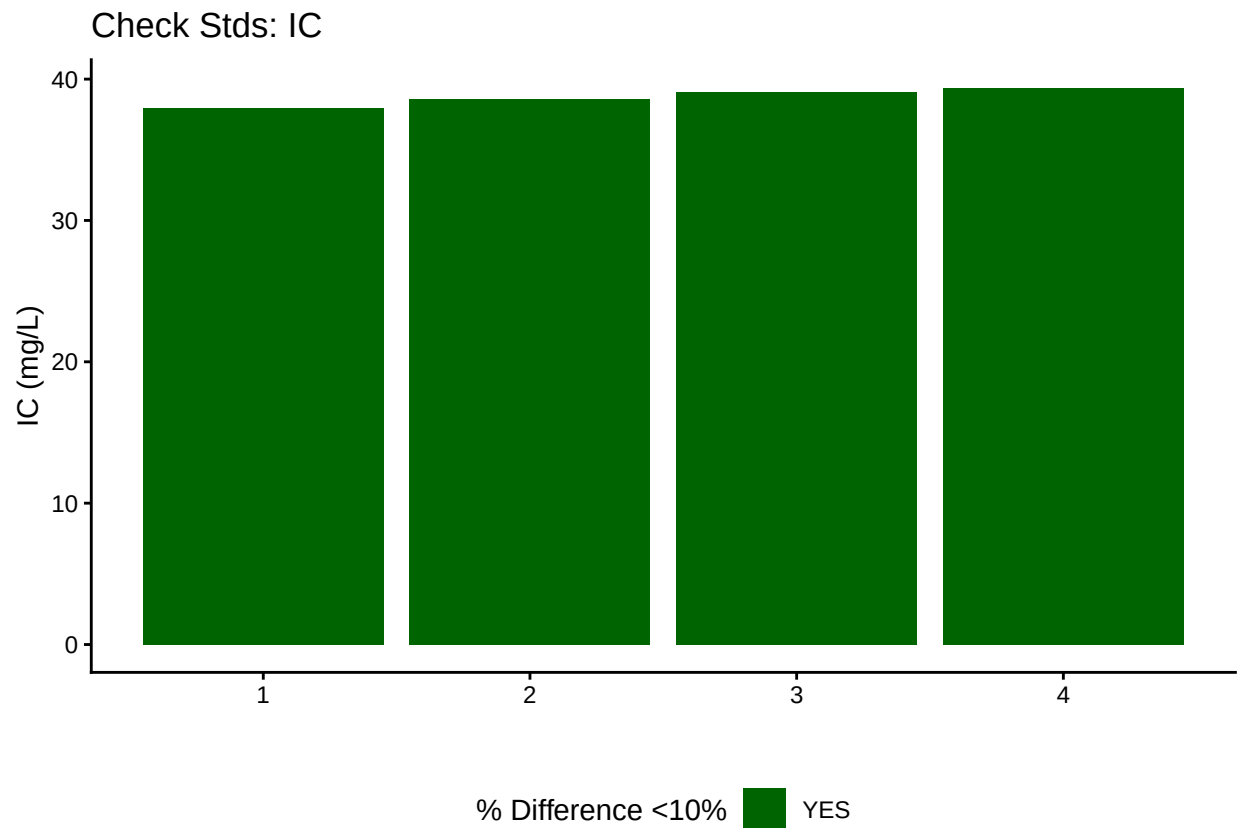
Assess Check Standards

```
## Assess the Check Standards
```

```
## New names:
```

```
## New names:
```

```
## * ' ' -> '...14'
```



```
## [1] ">60% of IC Check Standards are within range of expected concentration"
```

Assess Blanks

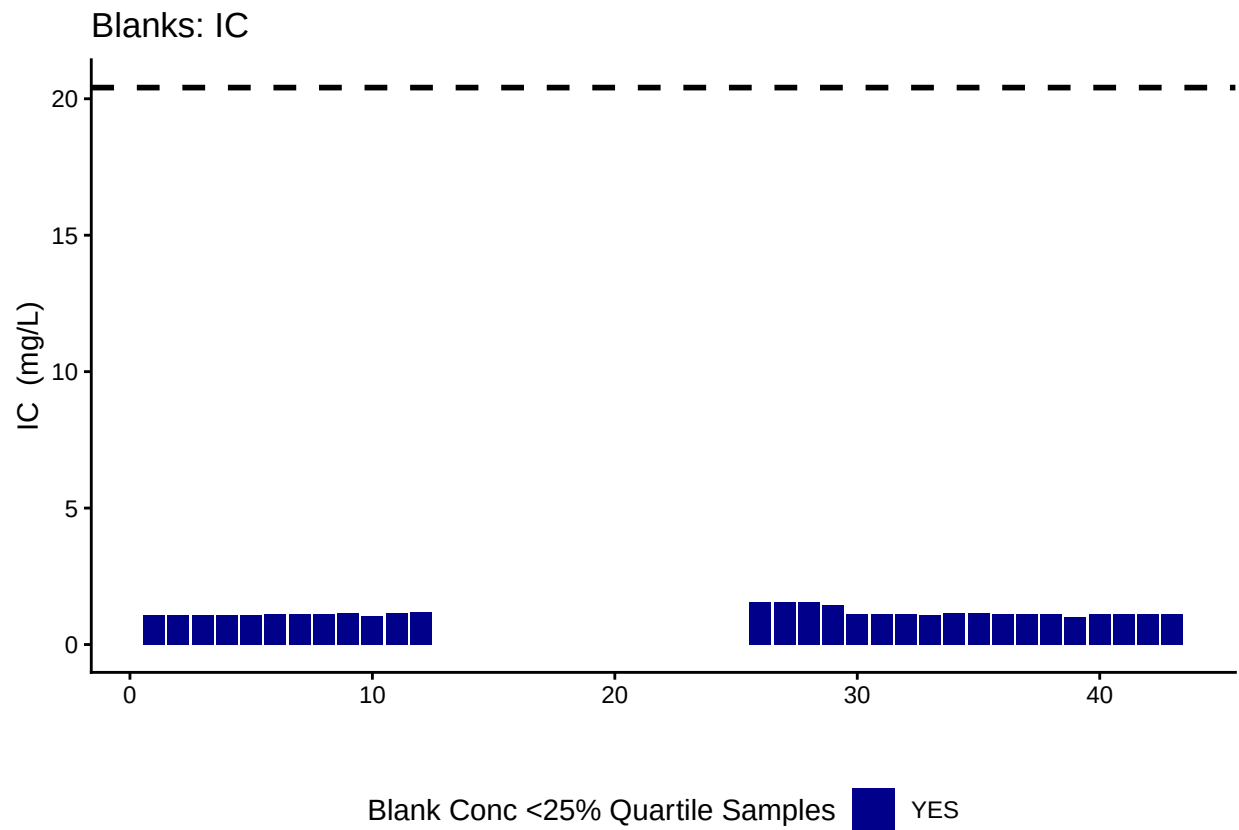
```
## Assess Blanks
```

```
## New names:
```

```
## New names:
```

```
## * ' ' -> '...14'
```

```
## [1] ">60% of Carbon Blank concentrations are lower 25% quartile of samples"
```

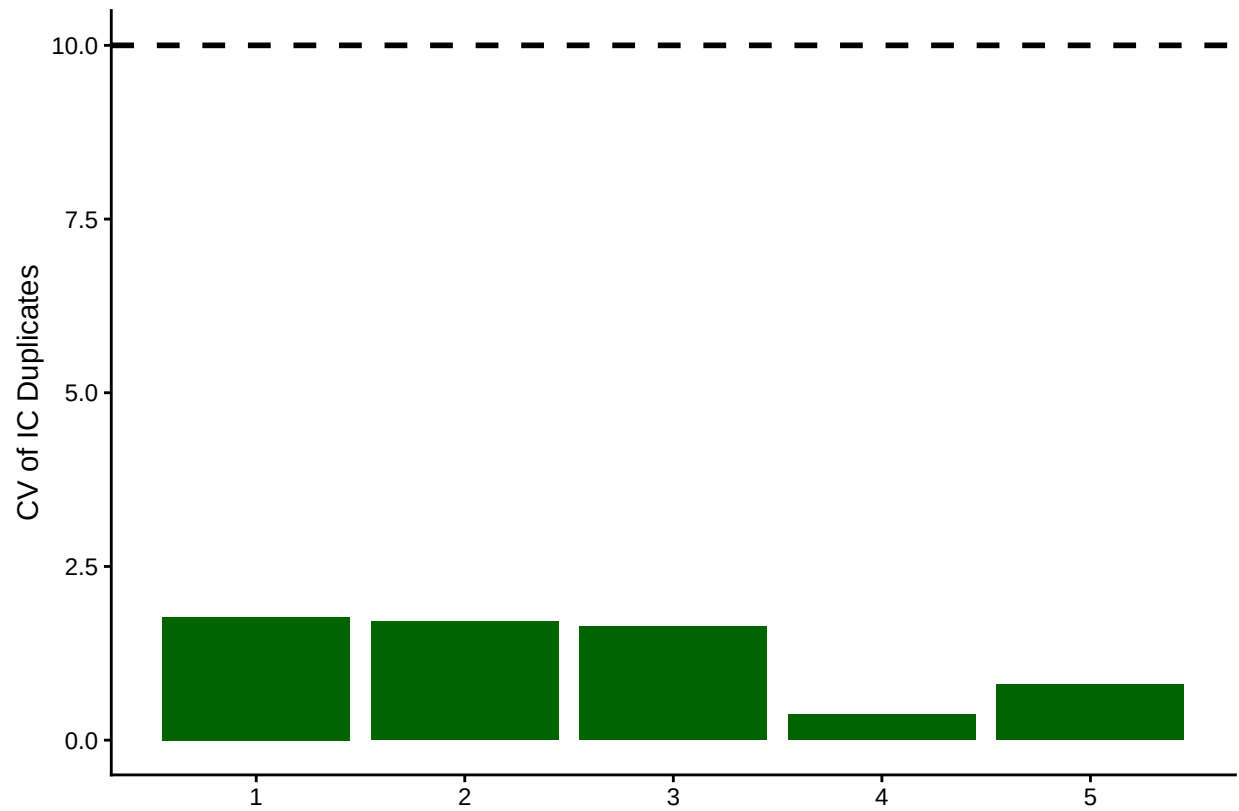


```
## carbon blanks:
```

```
## [1] 1.164467
```

Assess Duplicates

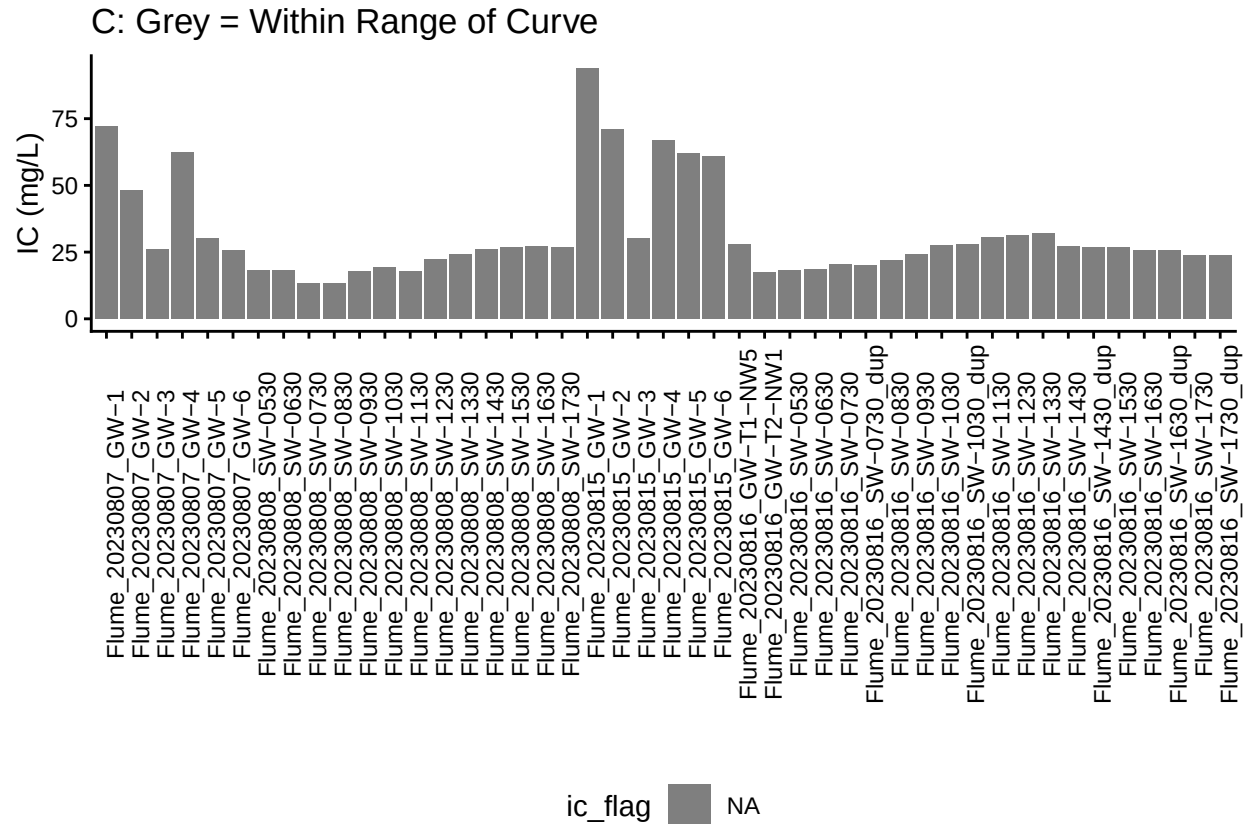
```
## Assess Duplicates
```



```
## [1] ">60% of Carbon Duplicates have a CV <10%"
```

Sample Flagging - Are samples Within the range of the curve?

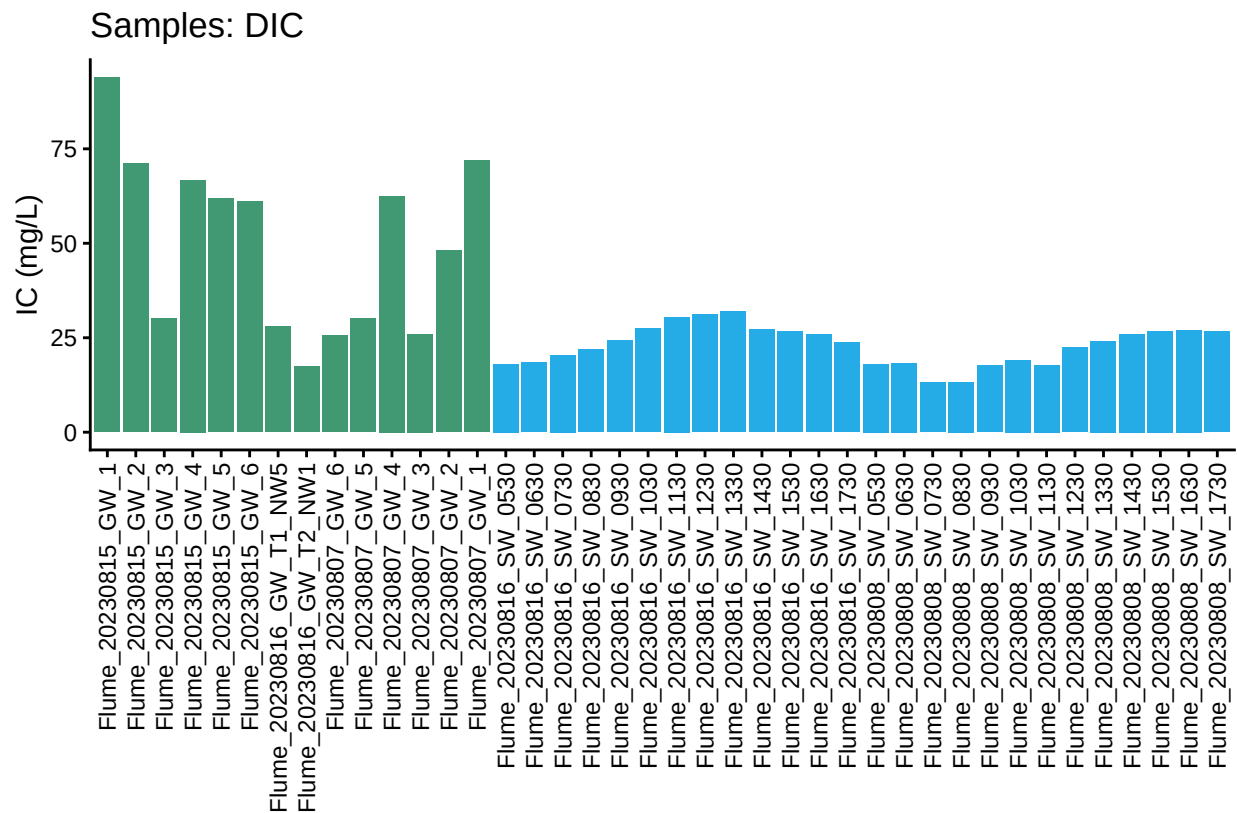
```
## Sample Flagging
```



Visualize Data by Plot - in 2023 SWH Upland = UPCON and Swamp = UP

```
## Visualize Data
```

```
## Warning in rbind(c("Flume", "20230815", "GW", "1"), c("Flume", "20230815", :
## number of columns of result is not a multiple of vector length (arg 1)
```



Convert data from mg/L to uMoles/L

Match metadata & merge info

Merge with Metadata

##	Survey_Type	Site	Date	Time_EDT	Time_EST	Sample_Type
## 1	Tidal Survey	Flume	8/8/2023	5:30	4:30	SW
## 2	Tidal Survey	Flume	8/8/2023	6:30	5:30	SW
## 3	Tidal Survey	Flume	8/8/2023	7:30	6:30	SW
## 4	Tidal Survey	Flume	8/8/2023	8:30	7:30	SW
## 5	Tidal Survey	Flume	8/8/2023	9:30	8:30	SW
## 6	Tidal Survey	Flume	8/8/2023	10:30	9:30	SW

##	Sample_ID	DO_mgL	pH	Salinity_ppt	Temp	DIC	DICex	SO4_H2S	NUTR	DOC
## 1	Flume_20230808_SW_0530	<NA>	NA		<NA>	NA	x	x	x	x
## 2	Flume_20230808_SW_0630	<NA>	NA		<NA>	NA	x	x	x	x
## 3	Flume_20230808_SW_0730	<NA>	NA		<NA>	NA	x	x	x	x
## 4	Flume_20230808_SW_0830	<NA>	NA		<NA>	NA	x	x	x	x
## 5	Flume_20230808_SW_0930	<NA>	NA		<NA>	NA	x	x	x	x
## 6	Flume_20230808_SW_1030	<NA>	NA		<NA>	NA	x	x	x	x

##	GHGs	TA	Radium_Vol_L	Notes
## 1	x	x	22.5	August: Neap Tide
## 2	x	x	22.5	August: Neap Tide
## 3	x	x	22.5	August: Neap Tide
## 4	x	x	22.5	August: Neap Tide

```
## 5      x      x      22.5 August: Neap Tide
## 6      x      x      22.5 August: Neap Tide
```

```
## All sample IDs are present in metadata.
```

Export Processed Data

```
## Export Processed Data
```

```
## # A tibble: 6 x 19
##   Project Site Replicate Sample_Type Sample_ID Date Time_EDT Time_EST ic_mgL
##   <chr>   <chr> <chr>      <chr>      <chr>   <chr> <chr>   <chr>   <dbl>
## 1 SERC Lat~ Flume 1      GW      Flume_20~ 8/15~ 11:30 10:30 93.9
## 2 SERC Lat~ Flume 2      GW      Flume_20~ 8/15~ 11:10 10:10 71.1
## 3 SERC Lat~ Flume 3      GW      Flume_20~ 8/15~ 10:50 9:50 30.2
## 4 SERC Lat~ Flume 4      GW      Flume_20~ 8/15~ 10:30 9:30 66.8
## 5 SERC Lat~ Flume 5      GW      Flume_20~ 8/15~ 10:11 9:11 61.9
## 6 SERC Lat~ Flume 6      GW      Flume_20~ 8/15~ 9:44 8:44 61.1
## # i 10 more variables: ic_uM <dbl>, ic_flag <chr>, Analysis_runtime <chr>,
## #   Run_notes <chr>, Field_notes <chr>, DO_mgL <chr>, pH <dbl>,
## #   Salinity_ppt <chr>, Temp <dbl>, Radium_Vol_L <dbl>
```

```
#end
```